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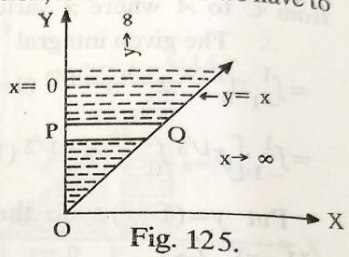
Department of Applied Mathematics & Humanities

Electronic Question Answer Bank

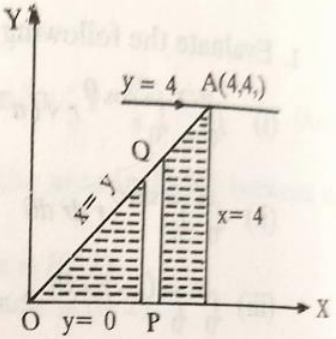
Course Name: Calculus and Vector **Course Code: 23GE1101**

UNIT-IV: MULTIPLE INTEGRALS AND THEIR APPLICATIONS

1	Unit 5	Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} (x^2+y^2)\,dy\,dx$ **Solution.** $\int_0^2 \int_0^{\sqrt{2x-x^2}} (x^2+y^2)\,dy\,dx$ Limits of $y = \sqrt{2x-x^2} \Rightarrow y^2 = 2x - x^2$ $\Rightarrow x^2 + y^2 - 2x = 0.$ ...(1) (1) represents a circle whose centre is (1, 0) and radius = 1. Lower limit of y is 0 *i.e.*, x-axis. Region of integration is upper half circle. Let us convert (1) into polar co-ordinates by putting $x = r\cos\theta,\ y = r\sin\theta$ $r^2 - 2r\cos\theta = 0 \Rightarrow r = 2\cos\theta$ Limits of r are 0 to $2\cos\theta$ Limits of θ are 0 to $\frac{\pi}{2}$ $\int_0^2 \int_0^{\sqrt{2x-x^2}} (x^2+y^2)\,dy\,dx = \int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} r^2\,(r\,d\theta\,dr)$ $= \int_0^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} r^3\,dr = \int_0^{\frac{\pi}{2}} d\theta \left[\frac{r^4}{4}\right]_0^{2\cos\theta}$ $= 4\int_0^{\frac{\pi}{2}} \cos^4\theta\,d\theta = 4\times\frac{3\times1\times\pi}{4\times2\times2} = \frac{3\pi}{4}$ Ans.	10	4	Low	Easy	CO-3	Numerical

2	Unit 5	Evaluate by changing the order of integration $\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dxdy$ **Solution.** From the limits of integration, it is clear that we have to integrate first w.r.t. y which varies from $y=x$ to $y=\infty$ and then w.r.t. x which varies from $x=0$ to $x=\infty$. The region of integration is shown shaded in figure. To change the order of integration we means to integrate first w.r.t. x and then w.r.t. y.  Fig. 125. To cover the region of integration, we draw a strip PQ parallel to X-axis where x varies from $x=0$ to $x=y$. Then this strip slides from O to infinite space where y varies from $y=0$ to $y=\infty$. $\therefore \int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dy\,dx=\int_0^{\infty}\int_0^{y}\frac{e^{-y}}{y}dy\,dx$ $=\int_0^{\infty}\left[\frac{e^{-y}}{y}\int_0^{y}dx\right]dy$ $=\int_0^{\infty}\frac{e^{-y}}{y}\left[x\right]_0^{y}dy$ $=\int_0^{\infty}\frac{e^{-y}}{y}\cdot y\,dy$ $=\int_0^{\infty}\frac{e^{-y}}{y}\cdot y\,dy$ $=\int_0^{\infty}e^{-y}dx$ $=\left[\frac{e^{-y}}{-1}\right]_0^{\infty}=[0+1]\quad\because \lim_{y\to\infty}e^{-y}=0$ $=1.$ **(Ans.)**	10	4	Low	Easy	CO-3	Numerical

3	Unit 5	Evaluate $\int_0^1 \int_0^y xye^{-x^2}\,dydx$ **Solution.** Here we are to first integrate w.r.t. x and then w.r.t. y. $\int_0^1 \int_0^y xye^{-x^2}\,dy\,dx = \int_0^1 \left[\int_0^y xy\,e^{-x^2}\,dx\right] dy$ $= \int_0^1 -\frac{y}{2}\left[\int_0^y -2x\,e^{-x^2}\,dx\right] dy$ $= \frac{1}{2}\int_0^1 -y\left[e^{-x^2}\right]_0^y\,dy$ $= \frac{1}{2}\int_0^1 -y\left[e^{-y^2}-1\right] dy$ $= \frac{1}{2}\int_0^1 \left[y - ye^{-y^2}\right] dy$ $= \frac{1}{2}\left[\frac{y^2}{2} + \frac{1}{2}e^{-y^2}\right]_0^1$ $= \frac{1}{4}\left[y^2 + e^{-y^2}\right]_0^1$ $= \frac{1}{4}\left[1 + \frac{1}{e} - 0 - 1\right] = \frac{1}{4e}$. (Ans.)	10	4	Low	Easy	CO-3	Numerical
4	Unit 5	Evaluate by changing the order of integration $\int_0^4 \int_y^4 \frac{x}{x^2+y^2} dxdy$	10	4	Low	Easy	CO-3	Numerical

		Solution. From the limits of integration, it is clear that we have to integrate first w.r.t. x which varies from $x=y$ to $x=4$ and then w.r.t. y which varies from $y=0$ to $y=4$. The region of integration is shown shaded in figure. To change the order of integration we means to integrate first w.r.t. y and then w.r.t. x.  Fig. 123. To cover the region of integration, we draw a strip PQ parallel to Y-axis where y varies from $y=0$ to $y=x$. Then this strip slides from O to A where x varies from $x=0$ to $x=4$. $\therefore \int_0^4 \int_y^4 \frac{x\,dx\,dy}{x^2+y^2} = \int_0^4 \int_0^x \frac{x\,dx\,dy}{x^2+y^2}$ $= \int_0^4 \left[x \int_0^x \frac{dy}{x^2+y^2} \right] dx$ $= \int_0^4 x \left[\frac{1}{x} \tan^{-1} \frac{y}{x} \right]_0^x dx$ $= \int_0^4 \frac{\pi}{4}\, dx$ (Ans.) $= \frac{\pi}{4} \left[x \right]_0^4 = \frac{\pi}{4} \times 4 = \pi$.						
5	Unit 5	Evaluate $\int_1^2 \int_0^1 \int_{-1}^1 (x^2 + y^2 + z^2)\,dxdydz$	10	4	Low	Easy	CO-3	Numerical

		Evaluate $\int_1^2 \int_0^1 \int_{-1}^1 (x^2+y^2+z^2)\, dx\, dy\, dz$ Sol. $I = \int_1^2 \int_0^1 \left[\int_{-1}^1 (x^2+y^2+z^2)\, dx \right] dy\, dz$ $= \int_1^2 \int_0^1 \left[\frac{x^3}{3} + (y^2+z^2)\, x \right]_{-1}^1 dy\, dz$ $= \int_1^2 \int_0^1 \left[\frac{1}{3} + (y^2+z^2) - \left(-\frac{1}{3} \right) - (y^2+z^2)\,(-1) \right] dy\, dz$ $= \int_1^2 \left[\int_0^1 \left\{ \frac{2}{3} + 2\left(y^2+z^2 \right) \right\} dy \right] dz$ $= \int_1^2 \left[\frac{2}{3} y + 2\left(\frac{y^3}{3} + z^2 y \right) \right]_0^1 dz$ $= \int_1^2 \left[\frac{2}{3} + 2\left(\frac{1}{3} + z^2 \right) \right] dz$						

		$=\int_1^2 \left(\frac{4}{3}+2z^2\right) dz$ $=\left[\frac{4}{3}z+\frac{2}{3}z^3\right]_1^2$ $=\frac{8}{3}+\frac{16}{3}-\frac{4}{3}-\frac{2}{3}$ $=\frac{18}{3}=6.$						
6	Unit 5	Evaluate $\iint e^{2x+3y} dxdy$ over the triangle bounded by x = 0, y = 0 and x + y = 1.	10	4	Low	Easy	CO-3	Numerical

Solution. The region of integration is the area bounded by the sides of the triangle OAB, shaded as shown in figure, where $A \equiv (1, 0)$ and $B \equiv (0, 1)$.

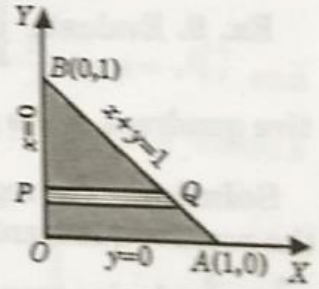
To find the limits of integration

Suppose we have to integrate first w.r.t. x and then w.r.t. y.

Draw a strip PQ parallel to X-axis in the region of integration where x varies from $x=0$ to $x=1-y$ and then PQ is slides from OA to B where y varies from $y=0$ to $y=1$,

$$\therefore \iint_R e^{2x+3y}\,dx\,dy = \int_{y=0}^{1}\int_{x=0}^{1-y} e^{2x+3y}\,dx\,dy$$

$$= \int_0^1 e^{3y}\left[\int_0^{1-y} e^{2x}\,dx\right]dy$$

		$= \frac{1}{2}\int_0^1 e^{3y}\left[e^{2(1-y)}-1\right]dy$ $= \frac{1}{2}\int_0^1 \left[e^{y+2}-e^{3y}\right]dy$ $= \frac{1}{2}\left[e^{y+2}-\frac{e^{3y}}{3}\right]_0^1$ $= \frac{1}{2}\left[e^3-\frac{e^3}{3}-e^2+\frac{1}{3}\right]$ $= \frac{1}{2}\left[\frac{2e^3}{3}-e^2+\frac{1}{3}\right]$ $= \frac{1}{6}\left[2e^3-3e^2+1\right]$ $= \frac{1}{6}(e-1)\left(2e^2-e-1\right) = \frac{1}{6}(e-1)^2(2e+1)$.(Ans.)  Fig. III.7						
7	Unit 5	Evaluate: $\int_0^1\int_{y^2}^{1}\int_0^{1-x} xdzdxdy$.	10	4	Low	Easy	CO-3	Numerical

Solution. First we are to integrate w.r.t. z, then w.r.t. x and finally w.r.t. y.

$$I=\int_0^1 \int_{y^2}^1 \left[x \int_0^{1-x} dz\right] dx\, dy$$

$$=\int_0^1 \int_{y^2}^1 x\left[z\right]_0^{1-x} dx\, dy$$

$$=\int_0^1 \int_{y^2}^1 x\,(1-x)\, dx\, dy$$

		$=\int_0^1 \left[\int_{y^2}^1 (x-x^2)\,dx\right] dy$ 373 $=\int_0^1 \left[\frac{x^2}{2}-\frac{x^3}{3}\right]_{y^2}^1 dy$ $=\int_0^1 \left(\frac{1}{2}-\frac{1}{3}-\frac{1}{2}y^4+\frac{1}{3}y^6\right) dy$ $=\int_0^1 \left(\frac{1}{6}-\frac{1}{2}y^4+\frac{1}{3}y^6\right) dy$ $=\left[\frac{1}{6}y-\frac{1}{10}y^5+\frac{1}{21}y^7\right]_0^1$ $=\frac{1}{6}-\frac{1}{10}+\frac{1}{21}$ $=\frac{35-21+10}{210}$ $=\frac{24}{210}=\frac{4}{35}$. (Ans.)						
8	Unit 5	Evaluate by changing the order of integration $\int_0^{\frac{\pi}{2}} \int_0^{a\cos\theta} r\sin\theta\, dr\, d\theta$	10	4	Low	Easy	CO-3	Numerical

Solution. From the limits of integration, it is clear that we have to integrate first w.r.t. r which varies from $r=0$ to $r=a\cos\theta$ [which is a circle with radius equal to $\frac{a}{2}$ and centre at the point $(a/2, 0)$] and then w.r.t. θ which varies from $\theta=0$ to $\theta=\pi/2$. The region of integration is shown shaded in figure.

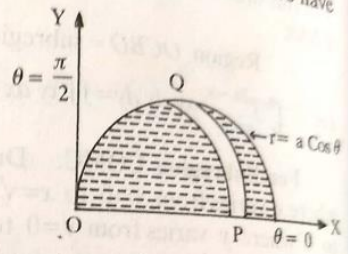


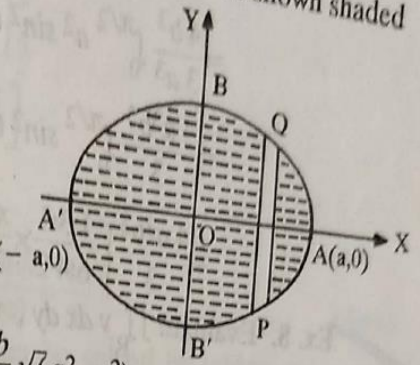
Fig. 130.

To change the order of integration we means to integrate first w.r.t. θ and then w.r.t. r.

To cover the region of integration, we draw a radial strip PQ starting from $\theta=0$ and ending on the circle $\theta=\cos^{-1}\frac{r}{a}$. Then this strip extending from $r=0$ to $r=a$.

$$\therefore \int_0^{\pi/2}\int_0^{a\cos\theta} r\sin\theta\, dr\, d\theta = \int_0^{a}\int_0^{\cos^{-1} r/a} r\sin\theta\, dr\, d\theta$$

$$= \int_0^{a}\left[r\int_0^{\cos^{-1} r/a}\sin\theta\, d\theta\right] dr$$

		369 $= \int_0^a r \left[-\cos\theta \right]_0^{\cos^{-1} r/a} dr$ $= \int_0^a -r\left(\frac{r}{a} - 1\right) dr$ $= \int_0^a \left(r - \frac{r^2}{a}\right) dr$ $= \left[\frac{r^2}{2} - \frac{r^3}{3a}\right]_0^a$ $= \frac{a^2}{2} - \frac{a^2}{3} = \frac{a^2}{6}$. (Ans.)						
9	Unit 5	Evaluate $\iint_R x^2 y^2 dxdy$, over the area bounded by ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ **Solution.** The region of integration $ABA'B'A$ is shown shaded in the figure. To cover the region of integration, we draw a strip $PQ \parallel$ to Y-axis where the strip PQ extends from a point P on the lower arc to the point Q on the upper arc of the ellipse *i.e.* y varies from $y = -\frac{b}{a}\sqrt{(a^2 - x^2)}$ to $y = \frac{b}{a}\sqrt{(a^2 - x^2)}$.  Fig. 117.	10	4	Low	Easy	CO-3	Numerical

		$= \frac{4b^3}{3a^3} \int_0^a x^2 (a^2 - x^2)^{3/2} dx$ $\because x^2 (a^2 - x^2)^{3/2}$ is an even function of x. Put $x = a \sin\theta$, so that $dx = a\cos\theta\, d\theta$ as $x \to 0$, $\theta \to 0$ and as $x \to a$, $\theta \to \frac{\pi}{2}$ $= \frac{4b^3}{3a^3} \int_0^{\pi/2} a^2 \sin^2\theta \cdot (a^2 - a^2\sin^2\theta)^{3/2} \cdot a\cos\theta\, d\theta$ $= \frac{4b^3}{3a^3} \int_0^{\pi/2} a^2 \sin^2\theta \cdot a^3 \cdot \cos^3\theta \cdot a\cos\theta\, d\theta$ $= \frac{4a^3b^3}{3} \int_0^{\pi/2} \sin^2\theta \cos^4\theta\, d\theta$ $= \frac{4a^3b^3}{3}\left[\frac{1\cdot3\cdot1}{6\cdot4\cdot2} \times \frac{\pi}{2}\right] = \frac{\pi a^3b^3}{24}$ (Ans.)						
10	Unit 5	Evaluate $\iint_R ydxdy$, where R is the region bounded by parabolas $y^2 = 4x$ and $x^2 = 4y$	12	4	Medium	Moderate	CO-3	Numerical

Solution. First we find the point of intersection of the parabolas

$$y^2=4x \quad \ldots(1)$$

and

$$x^2=4y \quad \ldots(2)$$

From (2), $y=\frac{x^2}{4}$. Putting this value of y in (1) then we have

$$\left(\frac{x^2}{4}\right)^2=4x$$

or $x^4-64x=0$

or $x(x^3-64)=0$

$\therefore \quad x=0,\ 4$

When $x=0$, (1) gives $y=0$

and when $x=4$, (1) gives $y=4$

$\therefore$ the points of intersection are $O(0,0)$ and $A(4,4)$.

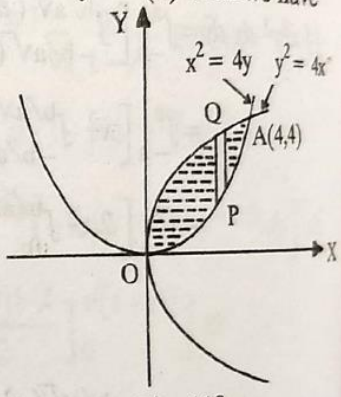


Fig. 118.

The region of integration R is shown shaded in the figure.

To cover the region of integration, we draw a strip $PQ \parallel$ to Y-axis where the strip PQ extends from a point P on the parabola $x^2=4y$ *i.e.* $y=\frac{x^2}{4}$ to the point Q on $y^2=4x$ *i.e.* $y=2\sqrt{x}$. Then the strip slides from O to A where x varies from $x=0$ to $x=4$.

		slides from O to A where x varies from $x=0$ to x $\therefore \iint_R y\,dx\,dy = \int_0^4 \int_{x^2/4}^{2\sqrt{x}} y\,dx\,dy$ $= \int_0^4 \left[\int_{x^2/4}^{2\sqrt{x}} y\,dy \right] dx$ $= \int_0^4 \left[\frac{y^2}{2} \right]_{x^2/4}^{2\sqrt{x}} dx$ $= \frac{1}{2} \int_0^4 \left[4x - \frac{x^4}{16} \right] dx$ $= \frac{1}{2} \left[2x^2 - \frac{x^5}{80} \right]_0^4$ $= \frac{1}{2} \left[32 - \frac{1024}{80} \right]$ $= \left[16 - \frac{512}{80} \right]$ $= \frac{1280-512}{80} = \frac{768}{80} = \frac{48}{5}$.						
11	Unit 5	Find the mass of the tetrahedron bounded by the co-ordinate plane and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$, the variable density $\rho = kxyz$	12	4	Mediu m	Moderate	CO-3	Numerical

Solution. The tetrahedron $OABC$ bounded by the planes $x=0,\ y=0,\ z=0$ and

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1.$$

To cover the whole volume of this tetrahedron

z varies from 0 to $c\left(1-\frac{x}{a}-\frac{y}{b}\right)$,

y varies from 0 to $b\left(1-\frac{x}{a}\right)$,

and x varies from 0 to a,

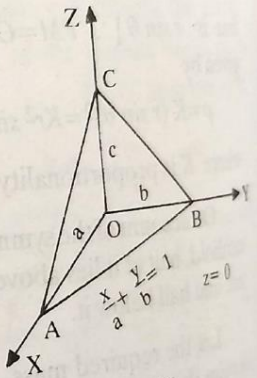


Fig. 156.

$\therefore$ Required mass

$$M=\int_0^a \int_0^{b(1-x/a)} \int_0^{c(1-x/a-y/b)} K\,xyz\,dz\,dy\,dx$$

$$=\int_0^a \int_0^{b(1-x/a)} K\,xy\left[\frac{z^2}{2}\right]_0^{c(1-x/a-y/b)} dy\,dx$$

$$=\int_0^a \int_0^{b(1-x/a)} \frac{K}{2}xy\left[c^2\left(1-\frac{x}{a}-\frac{y}{b}\right)^2\right] dy\,dx$$

$$=\int_0^a \left\{\frac{Kc^2}{2}x\int_0^{b(1-x/a)}\left[\left(1-\frac{x}{a}\right)^2 y-2\left(1-\frac{x}{a}\right)\frac{y^2}{b}+\frac{y^3}{b^2}\right] dy\right\} dx$$

$$=\int_0^a \frac{Kc^2}{2}x\left[\left(1-\frac{x}{a}\right)^2\cdot\frac{y^2}{2}-2\left(1-\frac{x}{a}\right)\cdot\frac{y^3}{3b}+\frac{y^4}{4b^2}\right]_0^{b(1-x/a)} dx$$

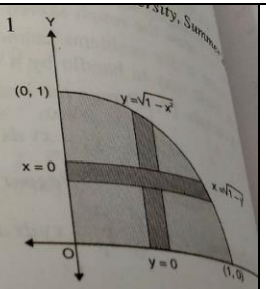
$$=\int_0^a \frac{Kc^2}{2}\cdot x\left[\frac{b^2}{2}\left(1-\frac{x}{a}\right)^4-\frac{2b^2}{3}\left(1-\frac{x}{a}\right)^4+\frac{b^2}{4}\left(1-\frac{x}{a}\right)^4\right] dx$$

$$=\int_0^a \frac{Kc^2}{2}\cdot\frac{b^2}{12}\cdot x\left(1-\frac{x}{a}\right)^4 dx$$

$$=\frac{Kb^2c^2}{24}\int_0^a x\left(1-\frac{x}{a}\right)^4 dx$$

		$= \frac{Kb^2c^2}{24} \int_0^a x\left(1-\frac{x}{a}\right)^4 dx$ $= \frac{Kb^2c^2}{24}\left[x \cdot \frac{\left(1-\frac{x}{a}\right)^5}{5\left(-\frac{1}{a}\right)} - (1) \cdot \frac{\left(1-\frac{x}{a}\right)^6}{5 \cdot 6 \cdot \frac{1}{a^2}}\right]_0^a$ $= \frac{Kb^2c^2}{24} \times \frac{a^2}{30}$ $= \frac{Ka^2b^2c^2}{720}$. **(Ans.)**						
12	Unit 5	Evaluate $\int_{-1}^{1}\int_{0}^{z}\int_{x-z}^{x+z}(x+y+z)dydxdz$ $= \int_{-1}^{1}\int_{0}^{z}\left[\int_{x-z}^{x+z}(x+y+z)\,dy\right]dx\,dz$ $= \int_{-1}^{1}\int_{0}^{z}\left[xy + \frac{y^2}{2} + zy\right]_{x-z}^{x+z} dx\,dz.$ $= \int_{-1}^{1}\int_{0}^{z}\left[x(x+z-x+z) + \frac{1}{2}\left((x+z)^2-(x-z)^2\right) + z(x+z-x+z)\right]dx$	12	4	Medium	Moderate	CO-3	Numerical

		$= \int_{-1}^{1}\int_{0}^{z}(2xz + 2z^3 + 2xz)\,dx\,dz.$ $= \int_{-1}^{1} kx^2z + 2z^2x + \frac{2x^2}{2}z \Big|_0^z dz.$ $\int_{-1}^{1} 2z(z^2) + 2z^2z.)\,dz.$ $\int_{-1}^{1} 2z(z^2) + 2z^2z.)\,dz.$ $= \int_{-1}^{1} 4z^3 dz = 0$ z^3 is an odd $\int^n$.						
13	Unit 5	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2\,dydx$	12	4	Medium	Moderate	CO-3	Numerical

		Solution. The region of integration is from $x = 0$ to $x = 1$ and $y = 0$ to $y = \sqrt{1 - x^2}$. Now change the order of integration with respect to x from $x = 0$ to $x = \sqrt{1 - y^2}$ and then $y = 0$ to $y = 1$. $\therefore \int_0^1 \int_0^{\sqrt{1-x^2}} y^2 \, dy \, dx$ $= \int_0^1 \int_0^{\sqrt{1-y^2}} y^2 \, dx \, dy = \int_0^1 y^2 [x]_0^{\sqrt{1-y^2}} \, dy$  $= \int_0^1 y^2 \sqrt{1 - y^2} \, dy$ Put $y = \sin\theta$ so that $dy = \cos\theta \, d\theta$ when $y = 0$, then $\theta = 0$ and $y = 1$, then $\theta = \frac{\pi}{2}$ $= \int_0^{\frac{\pi}{2}} \sin^2\theta \cos\theta \cos\theta \, d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{2\sin\theta\cos\theta}{2}\right)^2 d\theta$ $= \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin^2 2\theta \, d\theta = \frac{1}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 4\theta) \, d\theta$ $= \frac{1}{8} \left(\theta - \frac{\sin 4\theta}{4}\right)_0^{\frac{\pi}{2}}$ $= \frac{1}{8} \left\{\left(\frac{\pi}{2} - 0\right) - \frac{1}{4}(\sin 2\pi - \sin 0)\right\} = \frac{\pi}{16}$.						
14	Unit 5	Evaluate $\iint r^3 dr d\theta$ over the area included between the circle r = 2 sin θ and r = 4 sin θ	12	4	Medium	Moderate	CO-3	Numerical

Ex 3. Evaluate $\iint$ ... between the circles $r=2\sin\theta$ and $r=4\sin\theta$.

Solution. Centre of the first circle is (0, 1).
Centre of the second circle is (0, 2).
Radius of the first circle is 1 and
Radius of the second circle is 2.

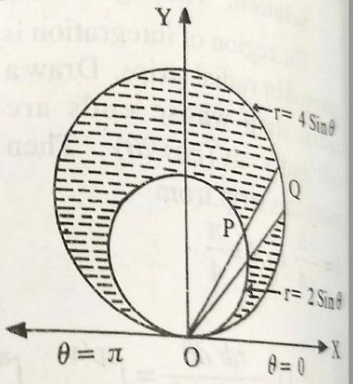


Fig.12.2.

The region of integration is shown shaded in the figure.

The region of integration is covered by radial strips. Draw a radial strip PQ whose ends are $r=2\sin\theta$ and $r=4\sin\theta$. Then rotate this strip from $\theta=0$ to $\theta=\pi$.

$$\therefore \iint_R r^3\,dr\,d\theta = \int_0^{\pi}\int_{2\sin\theta}^{4\sin\theta} r^3\,dr\,d\theta$$

$$= \int_0^{\pi}\left[\int_{2\sin\theta}^{4\sin\theta} r^3\,dr\right] d\theta$$

		$= \int_0^{\pi} \left[\frac{r^4}{4} \right]_{2\sin\theta}^{4\sin\theta} d\theta$ $= \frac{1}{4} \int_0^{\pi} [256 \sin^4\theta - 16 \sin^4\theta] d\theta$ $= \frac{1}{4} \int_0^{\pi} 240 \sin^4\theta \, d\theta$ $= 60 \times 2 \int_0^{\pi/2} \sin^4\theta \, d\theta$ $= 120 \times \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2}$ $= \frac{45\pi}{2}$						
15	Unit 5	Evaluate $\int_0^{\pi} \int_0^{a(1-\cos\theta)} r^2 \sin\theta \, dr \, d\theta$	12	4	Medium	Moderate	CO-3	Numerical

		Solution. $\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2 \sin\theta \, d\theta \, dr$ $= \int_0^{\pi}\left(\frac{r^3}{3}\right)_0^{a(1-\cos\theta)} \sin\theta \, d\theta = \frac{a^3}{3}\int_0^{\pi}\sin\theta(1-\cos\theta)^3 \, d\theta$ [Put, $1-\cos\theta = t$, $\therefore \sin\theta \, d\theta = dt$] $= \frac{a^3}{3}\int_0^2 t^3 \, dt = \frac{a^3}{3}\left(\frac{t^4}{4}\right)_0^2 = \frac{1}{12}a^3(16) = \frac{4}{3}a^3.$ Ans.						
16	Unit 5	Evaluate $\iiint_R (x-2y+z)dxdydz$, Where R: $0 \le x \le 1$; $0 \le y \le x^2$; $0 \le z \le x+y$ **Solution.** $\int\int\int_R (x-2y+z)\,dx\,dy\,dz$ $= \int_0^1 dx \int_0^{x^2} dy \int_0^{x+y} (x-2y+z)\,dz = \int_0^1 dx \int_0^{x^2} dy \left(xz-2yz+\frac{z^2}{2}\right)_0^{x+y}$ $= \int_0^1 dx \int_0^{x^2} dy \left(x^2+xy-2xy-2y^2+\frac{(x+y)^2}{2}\right) = \int_0^1 dx \int_0^{x^2} dy \left(x^2-xy-2y^2+\frac{x^2}{2}+\frac{y^2}{2}+xy\right)$ $= \int_0^1 dx \int_0^{x^2} dy \left(\frac{3x^2}{2}-\frac{3y^2}{2}\right) = \frac{3}{2}\int_0^1 dx \int_0^{x^2}(x^2-y^2)\,dy = \frac{3}{2}\int_0^1 dx \left(x^2y-\frac{y^3}{3}\right)_0^{x^2}$ $= \frac{3}{2}\int_0^1 dx\left(x^4-\frac{x^6}{3}\right) = \frac{3}{2}\left[\frac{x^5}{5}-\frac{x^7}{21}\right]_0^1 = \frac{3}{2}\left(\frac{1}{5}-\frac{1}{21}\right) = \frac{8}{35}$	12	4	Medium	Moderate	CO-3	Numerical
17	Unit 5	Find the mass of the plate which is inside the circle r = 2a sin θ and outside the circle r = a, if density varies as the distance from the pole.	12	4	Medium	Moderate	CO-3	Numerical

Solution. The portion of the plate is shown shaded in the figure.

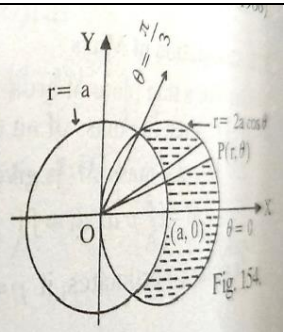
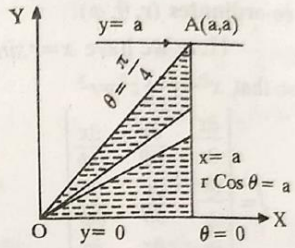


Fig. 15.4.

Since the distance of any point $P(r, \theta)$ from the pole is r, the density at $P(r, \theta)$ is given by $\rho = Kr$, where K is proportionality constant.

		 On account of the symmetry of the plate, half of it lies above the initial line and half below it. Let the required mass be M which is twice the mass above the initial line. For the region above the initial line, r varies from a to $2a\cos\theta$ and θ varies from 0 to $\frac{\pi}{3}$. $\therefore\ M=2\int_0^{\pi/3}\int_a^{2a\cos\theta}\rho r\,dr\,d\theta$ $=2\int_0^{\pi/3}\int_a^{2a\cos\theta}Kr^2\,dr\,d\theta$ $=2\int_0^{\pi/3}K\left[\frac{r^3}{3}\right]_a^{2a\cos\theta}d\theta$ $=\frac{2K}{3}\int_0^{\pi/3}(8a^3\cos^3\theta-a^3)\,d\theta$ $=\frac{2Ka^3}{3}\int_0^{\pi/3}\left[8\left(\frac{\cos 3\theta+3\cos\theta}{4}\right)-1\right]d\theta$ $\because\ \cos 3\theta=4\cos^3\theta-3\cos\theta$ $=\frac{2Ka^3}{3}\left[2\left(\frac{\sin 3\theta}{3}+3\sin\theta\right)-\theta\right]_0^{\pi/3}$ $=\frac{2Ka^3}{3}\left[6\cdot\frac{\sqrt{3}}{2}-\frac{\pi}{3}\right]$ $=\frac{2Ka^3}{3}$ (Ans.)						

		$= \frac{2Ka^3}{3}\left[6 \cdot \frac{\sqrt{3}}{2} - \frac{\pi}{3}\right]$ $= \frac{2Ka^3}{3}\left(3\sqrt{3} - \frac{\pi}{3}\right)$. (Ans.)						
18	Unit 5	Evaluate $\int_0^a \int_y^a \frac{x^2}{(x^2+y^2)^{3/2}} dy dx$,by changing it to polar Coordinates **Solution.** From the limits of integration, it is clear that we have to integrate first w.r.t. x which varies from $x=y$ to $x=a$ and then w.r.t. y which varies from $y=0$ to $y=a$. The region of integration is shown shaded in figure. Changing cartesian co-ordinates (x, y) to polar co-ordinates (r, θ) by putting $x=r\cos\theta$, $y=r\sin\theta$ so that $x^2+y^2=r^2$ and $dx\,dy=r\,dr\,d\theta$.  Fig. 133. The region of integration is covered by radial strips. Draw a radial strip whose ends are $r=0$ and $r=\frac{a}{\cos\theta}$. Then rotate this strip from $\theta=0$ to $\frac{\pi}{4}$. $\therefore \int_0^a \int_y^a \frac{x^2\,dy\,dx}{(x^2+y^2)^{3/2}} = \int_0^{\pi/4} \int_0^{a/\cos\theta} \frac{r^2\cos^2\theta}{(r^2)^{3/2}} r\,dr\,d\theta$	12	4	Medium	Difficult	CO-3	Numerical

		$= \int_0^{\pi/4} \left[\cos^2\theta \int_0^{a/\cos\theta} dr \right] d\theta$ $= \int_0^{\pi/4} \cos^2\theta \left[r \right]_0^{a/\cos\theta} d\theta$ $= \int_0^{\pi/4} \cos^2\theta \cdot \frac{a}{\cos\theta} d\theta$ $= a \int_0^{\pi/4} \cos\theta \, d\theta$ $= a \left[\sin\theta \right]_0^{\pi/4}$ $= a \sin\frac{\pi}{4} = \frac{a}{\sqrt{2}}$. (Ans.)						
19	Unit 5	Let OA be the diameter of a semicircular disc,the density at any point varies as its distance from O. Find the position of center of gravity , given OA = a.	12	4	Mediu m	Difficult	CO-3	Numerical

Solution. Since the distance of any point $P(r, \theta)$ O is r, then density at $P(r, \theta)$ is given by $\rho = Kr$.

From fig. $\cos\theta = \frac{OP}{OA} = \frac{r}{a}$

$\Rightarrow \quad r = a\cos\theta$

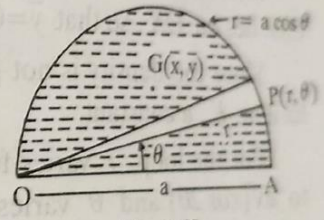


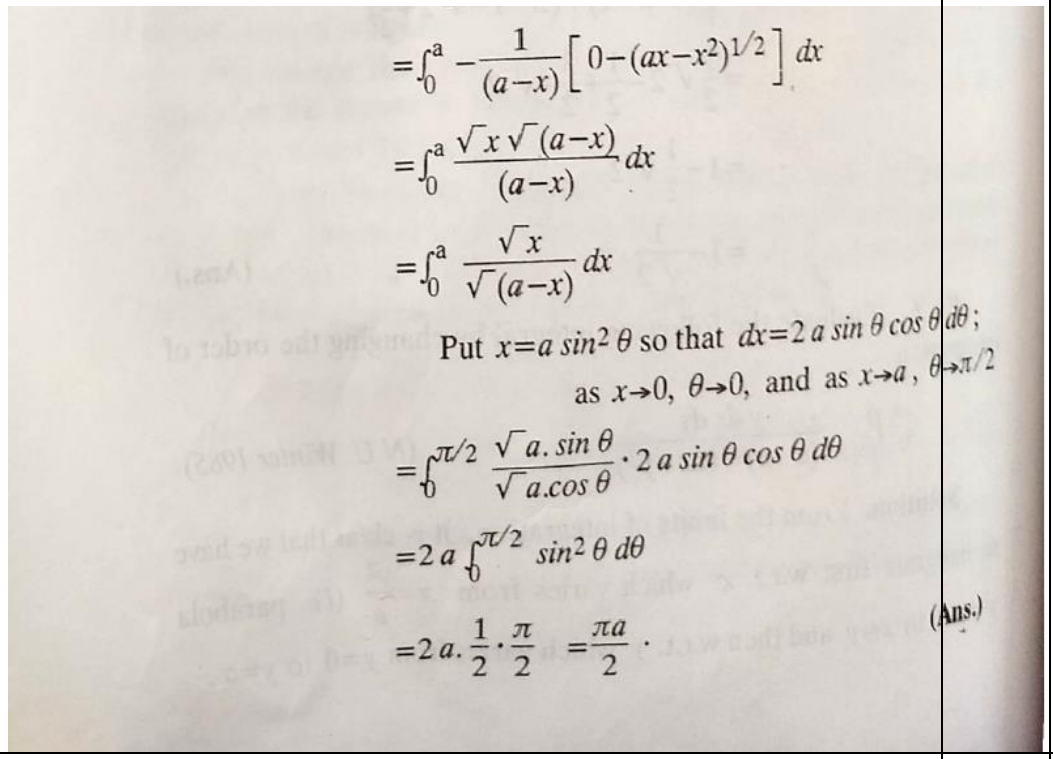
Fig. 159.

For a semicircular disc, r varies from O to $a\cos\theta$ and θ varies from O to $\pi/2$.

If $G(\bar{x}, \bar{y})$ is the C.G. of the disc, then

$$\bar{x} = \frac{\int_0^{\pi/2}\int_0^{a\cos\theta} r\cos\theta \cdot \rho\, r\, dr\, d\theta}{\int_0^{\pi/2}\int_0^{a\cos\theta} \rho\, r\, dr\, d\theta}, \quad \bar{y} = \frac{\int_0^{\pi/2}\int_0^{a\cos\theta} r\sin\theta \cdot \rho\, r\, dr\, d\theta}{\int_0^{\pi/2}\int_0^{a\cos\theta} \rho\, r\, dr\, d\theta}$$

		$= \dfrac{\int_0^{\pi/2}\int_0^{a\cos\theta} r\cos\theta\, Kr\cdot r\,dr\,d\theta}{\int_0^{\pi/2}\int_0^{a\cos\theta} Kr\cdot r\,dr\,d\theta}$, $= \dfrac{\int_0^{\pi/2}\int_0^{a\cos\theta} r\sin\theta\cdot Kr\cdot r\,dr\,d\theta}{\int_0^{\pi/2}\int_0^{a\cos\theta} Kr\cdot r\,dr\,d\theta}$ $= \dfrac{\int_0^{\pi/2}\cos\theta\left[\frac{r^4}{4}\right]_0^{a\cos\theta}d\theta}{\int_0^{\pi/2}\left[\frac{r^3}{3}\right]_0^{a\cos\theta}d\theta}$, $= \dfrac{\int_0^{\pi/2}\sin\theta\left[\frac{r^4}{4}\right]_0^{a\cos\theta}d\theta}{2a^3/9}$ $= \dfrac{\frac{a^4}{4}\int_0^{\pi/2}\cos^5\theta\,d\theta}{\frac{a^3}{3}\int_0^{\pi/2}\cos^3\theta\,d\theta}$, $= \dfrac{\frac{a^4}{4}\int_0^{\pi/2}\sin\theta\cdot\cos^4\theta\,d\theta}{2a^3/9}$ $= \dfrac{\frac{a^4}{4}\cdot\frac{4\cdot2}{5\cdot3\cdot1}}{\frac{a^3}{3}\cdot\frac{2}{3\cdot1}}$, $= \dfrac{\left(-\frac{a^4}{4}\right)\left[\frac{\cos^5\theta}{5}\right]_0^{\pi/2}}{2a^3/9}$ $= \dfrac{2a^4/15}{2a^3/9}$, $= -\dfrac{a^4}{4}\left(0-\dfrac{1}{5}\right)\times\dfrac{9}{2a^3}$ $= \dfrac{3a}{5}$, $= \dfrac{a^4}{20}\times\dfrac{9}{2a^3} = \dfrac{9a}{40}$. Hence the required C.G. is $G\left(\dfrac{3a}{5}, \dfrac{9a}{40}\right)$. (Ans.)						
20	Unit 5	Evaluate by changing the order of integration $\int_0^a \int_{\frac{y^2}{a}}^{y} \frac{y}{(a-x)\sqrt{ax-y^2}}dxdy$ **Solution.** From the limits of integration, it is clear that we have to integrate first w.r.t. x which varies from $x=\frac{y^2}{a}$ (i.e. parabola $y^2=ax$) to $x=y$ and then w.r.t. y which varies from $y=0$ to $y=a$.	12	4	Medium	Difficult	CO-3	Numerical

		 $= \int_0^a -\frac{1}{(a-x)}\left[0-(ax-x^2)^{1/2}\right] dx$ $= \int_0^a \frac{\sqrt{x}\sqrt{(a-x)}}{(a-x)} dx$ $= \int_0^a \frac{\sqrt{x}}{\sqrt{(a-x)}} dx$ Put $x = a\sin^2\theta$ so that $dx = 2a\sin\theta\cos\theta\, d\theta$; as $x \to 0$, $\theta \to 0$, and as $x \to a$, $\theta \to \pi/2$ $= \int_0^{\pi/2} \frac{\sqrt{a}.\sin\theta}{\sqrt{a}.\cos\theta} \cdot 2a\sin\theta\cos\theta\, d\theta$ $= 2a\int_0^{\pi/2} \sin^2\theta\, d\theta$ $= 2a.\frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi a}{2}$. (Ans.)						
21	Unit 5	Evaluate by changing the order of integration $\int_0^a \int_{x^2/a}^{2a-x} xydxdy$	12	4	Medium	Difficult	CO-3	Numerical

Solution. From the limits of integration, it is clear that we have to integrate first w.r.t. y which varies from $y=\frac{x^2}{a}$ (*i.e.* parabola $x^2=ay$) to $y=2a-x$ [which is st. line intersecting X-axis in $A(2a, 0)$ and Y-axis in $B(0, 2a)$] and then w.r.t. x which varies from $x=0$ to $x=a$. The region of integration (*i.e.* region $OCBO$) is shown shaded in figure.

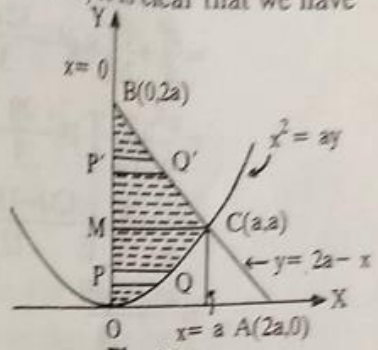


Fig. 129.

To change the order of integration we means to integrate first w.r.t. x and then w.r.t. y.

To cover the region of integration, we draw a strip PQ parallel to X-axis. But in doing so we see that strip extends over different curves, hence we divide the region of integration into two subregions. For this draw $CM \perp OY$. Therefore two subregions are $COMC$ and $CBMC$.

$\therefore$ Region $OCBO$ = subregion $COMC$ + subregion $CBMC$

i.e. $$\int_0^a \int_{x^2/a}^{2a-x} xy\,dx\,dy = \iint xy\,dx\,dy + \iint xy\,dx\,dy \quad \ldots(1)$$

For Subregion COMC. Draw a strip PQ parallel to X-axis where x varies from $x=0$ to $x=\sqrt{(ay)}$. Then this strip slides from O to C where y varies from $y=0$ to $y=a$.

For subregion CBMC. Draw a strip $P'Q'$ parallel to X-axis where x varies from $x=0$ to $x=2a-y$. Then this strip slides from C to B where y varies from $y=a$ to $y=2a$.

		∴ (1) becomes $\int_0^a \int_{x^2/a}^{2a-x} xy\,dx\,dy = \int_0^a \int_0^{\sqrt{(ay)}} xy\,dx\,dy + \int_a^{2a} \int_0^{2a-y} xy\,dx\,dy$ $= \int_0^a \left[y \int_0^{\sqrt{(ay)}} x\,dx \right] dy + \left[\int_a^{2a} \left[y \int_0^{2a-y} x\,dx \right] dy \right.$ $= \int_0^a y \left[\frac{x^2}{2} \right]_0^{\sqrt{(ay)}} dy + \int_a^{2a} y \left[\frac{x^2}{2} \right]_0^{2a-y} dy$ $= \frac{1}{2} \int_0^a y\,(ay)\,dy + \frac{1}{2} \int_a^{2a} y\,(2a-y)^2\,dy$						

		$=\frac{a}{2}\int_0^a y^2\,dy+\frac{1}{2}\int_a^{2a}(4a^2y-4ay^2+y^3)\,dy$ $=\frac{a}{2}\left[\frac{y^3}{3}\right]_0^a+\frac{1}{2}\left[2a^2y^2-\frac{4a}{3}y^3+\frac{y^4}{4}\right]_a^{2a}$ $=\frac{a^4}{6}+\frac{1}{2}\left[8a^4-\frac{32a^4}{3}+4a^4-2a^4+\frac{4}{3}a^4-\frac{1}{4}a^4\right]$ $=\frac{a^4}{6}+\frac{a^4}{2}\left[10-\frac{28}{3}-\frac{1}{4}\right]$ $=\frac{a^4}{6}+\frac{a^4}{2}\left[\frac{120-112-3}{12}\right]$ $=\frac{a^4}{6}+\frac{5a^4}{24}$ $=\frac{9a^4}{24}=\frac{3}{8}a^4.$						
22	Unit 5	Evaluate by changing the order of integration $\int_0^b\int_{y-b}^{2y} xydxdy$	12	4	Medium	Difficult	CO-3	Numerical

Solution. From the limits of integration, it is clear that we have to integrate firs w.r.t. x which varies from $x=y-b$ [which is a st. line intersecting X-axis in $(-b, 0)$ and Y-axis in $(0, b)$] to $x=2y$ and then w.r.t. y which varies from $y=0$ to $y=b$. The region of integration (*i.e.* region $OABCO$) is shown shaded in figure.

(N.U. Summer 1985)

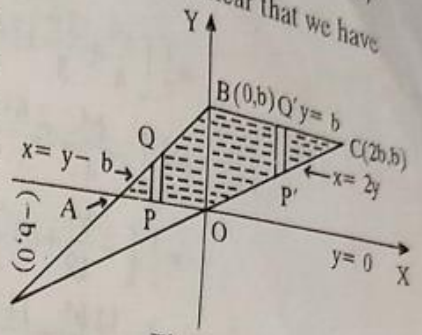


Fig. 126.

To change the order of integration we means to integrate first w.r.t. y and then w.r.t. x.

To cover the region of integration, we draw a strip PQ parallel to Y-axis. But in doing so we see that strip extends over different curves, hence we divide the region of integration into two subregions (i) subregion $OABO$ (ii) subregion $OBCO$,

$\therefore$ Region OABCO = subregion $OABO$ + subregion $OBCO$

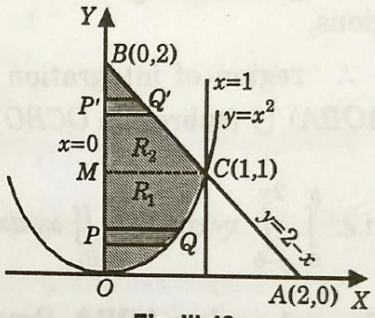
i.e. $$\int_0^b \int_{y-b}^{2y} xy\, dx\, dy = \iint xy\, dx\, dy + \iint xy\, dx\, dy \quad \ldots(1)$$

For subregion OABO : Draw a strip PQ parallel to Y-axis where varies from $y=0$ to $y=x+b$. Then this strip slides from A to B here x varies from $x=-b$ to $x=0$.

For subregion OBCO : Draw a strip $P'Q'$ parallel to Y-axis here y varies from $y=\frac{x}{2}$ to $y=b$. Then this strip slides from O to where x varies from $x=0$ to $x=2b$.

$\therefore$ (1) becomes

$$\int_0^b \int_{y-b}^{2y} xy\, dx\, dy = \int_{-b}^{0} \int_{0}^{x+b} xy\, dx\, dy + \int_0^{2b} \int_{x/2}^{b} xy\, dx\, dy$$

23	Unit 5	Evaluate by changing the order of integration $\int_0^1 \int_{x^2}^{2-x} xydxdy$ **Solution.** From the given limits of integration, it is clear that we have to integrate first w.r.t. y where y varies from $y=x^2$ to $y=2-x$ which is a st. line meets X-axis in $(2, 0)$ and Y-axis in $(0, 2)$ and then w.r.t. x where x varies from $x=0$ to $x=1$. The region of integration bounded by four curves is shaded as shown in figure.  **Fig. III.43** In changing the order of integration, we integrate first w.r.t. x and then w.r.t. y. Draw a strip parallel to X-axis in the region of integration. But in dong so we observe that ends of the strip lie on the different curves, this indicates that the region of integration is divided into two sub-regions. For that draw $CM \perp OY$, so that region of integration $OABMO$ =(subregion $OCMO$) $\cup$ (subregion $CBMC$) $$i.e. \int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx = \iint_{R_1} xy\,dy\,dx + \iint_{R_2} xy\,dy\,dx \quad ...(1)$$	12	4	Medium	Difficult	CO-3	Numerical

For subregion OCMO. Draw a strip PQ parallel to X-axis in this region where x varies from $x=0$ to $x=\sqrt{y}$ and strip PQ is then slides from O to C where y varies from $y=0$ to $y=1$.

For subregion CBMC. Draw a strip $P'Q'$ parallel to X-axis in this region where x varies from $x=0$ to $x=2-y$ and strip $P'Q'$ is then slides from C to B where y varies from $y=1$ to $y=2$,

$\therefore$ (1) becomes

$$\int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx = \int_{y=0}^{1} \int_{x=0}^{\sqrt{y}} xy\,dy\,dx + \int_{y=1}^{2} \int_{x=0}^{2-y} xy\,dy\,dx$$

$$= \int_0^1 y \left[\frac{x^2}{2}\right]_0^{\sqrt{y}} dy + \int_1^2 y \left[\frac{x^2}{2}\right]_0^{2-y} dy$$

$$= \frac{1}{2}\left\{\int_0^1 y^2\,dy + \int_1^2 y(2-y)^2\,dy\right\}$$

		$=\frac{1}{2}\left\{\left[\frac{y^3}{3}\right]_0^1+\left[2y^2-4\frac{y^3}{3}+\frac{y^4}{4}\right]_1^2\right\}$ $=\frac{1}{2}\left\{\frac{1}{3}+\left[8-\frac{32}{3}+4-2+\frac{4}{3}-\frac{1}{4}\right]\right\}$ $=\frac{1}{2}\left\{10-9-\frac{1}{4}\right\}=\frac{1}{2}\times\frac{3}{4}=\frac{3}{8}.$ (An						
24	Unit 5	$\iint_R x^2\,dxdy$ **where R is the region in the first quadrant bounded by hyperbola xy = 16 and line y=x , y=o,x=8**	**12**	**4**	**Medium**	**Difficult**	CO-3	**Numerical**

Solution. First we find the point of intersection of the hyperbola

$$xy=16 \quad \ldots(1)$$

and the line

$$y=x \quad \ldots(2)$$

Putting $y=x$ in (1) we have

$$x^2=16 \qquad \therefore \quad x=4$$

When $x=4$, (2) gives $y=4$.

$\therefore$ Its co-ordinates are (4, 4)

Next to find the point of intersection of (1) and the line

$$x=8 \quad \ldots(3)$$

Putting $x=8$ in (1), we get $y=2$.

Let B be the point of intersection of (1) and (3)

$\therefore$ Its co-ordinates are (8, 2).

The region of integration R (*i.e. ONBAO*) is shown shaded in the figure.

To cover the region of integration, draw a strip PQ parallel to Y-axis but in doing so we see that strip PQ extends over different curves

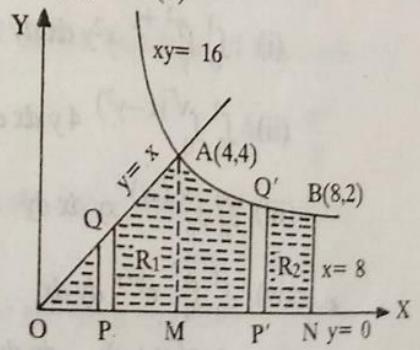


Fig. 119.

$$= \left[\frac{x^4}{4}\right]_0^4 + 16\left[\frac{x^2}{2}\right]_4^8$$

$$= \frac{1}{4}(256) + 8\,(64-16)$$

$$= 64 + 384 = 448.$$

Ans.

		hence we divide the region of integration R into two subregions R_1 and R_2. For this from A draw AM perpendicular on X-axis. $\therefore$ Region $ONBAO$=Subregion $OMAO$+Subregion $MABNM$ *i.e.* $\iint_R x^2\,dx\,dy=\iint_{R_1} x^2\,dx\,dy+\iint_{R_2} x^2\,dx\,dy$ ...(4) **For subregion R_1 (OMAO).** Draw a strip PQ parallel to Y-axis, y varies from $y=0$ to $y=x$. Then the strip PQ slides from O to A where x varies from $x=0$ to $x=4$. **For subregion R_2 (MABNM).** Draw a strip $P'Q'$ parallel to Y-axis, y varies from $y=0$ to $y=\frac{16}{x}$ $\left(\because xy=16 \therefore y=\frac{16}{x}\right)$. Then the strip $P'Q'$ slides from A to N where x varies from $x=4$ to $x=8$ $\therefore$ (4) becomes $\iint_R x^2\,dx\,dy=\int_0^4\left[\int_0^x x^2\,dy\right]dx+\int_4^8\left[\int_0^{16/x} x^2\,dy\right]dx$ $=\int_0^4 x^2\left[y\right]_0^x dx+\int_4^8 x^2\left[y\right]_0^{16/x} dx$ $=\int_0^4 x^3\,dx+\int_4^8 16x\,dx$ $=\left[\frac{x^4}{4}\right]_0^4+16\left[\frac{x^2}{2}\right]_4^8$ $=\frac{1}{4}(256)+8(64-16)$ $=64+384=448.$ (Ans.)						
25	Unit 5	Evaluate $\int_0^{\pi/2}\int_0^{asin\theta}\int_0^{(a^2-r^2)/a} rdrd\theta dz$	12	4	Medium	Difficult	CO-3	Numerical

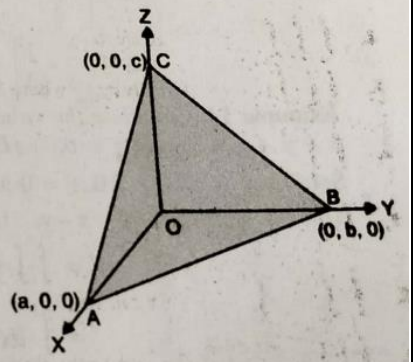
		Solution. First we are to integrate w.r.t. z, then w.r.t. r and finally w.r.t. θ. $I=\int_0^{\pi/2}\int_0^{a\sin\theta}\left[r\int_0^{(a^2-r^2)/a} dz\right] dr\, d\theta$ $=\int_0^{\pi/2}\int_0^{a\sin\theta} r\left[z\right]_0^{(a^2-r^2)/a} dr\, d\theta$ $=\int_0^{\pi/2}\int_0^{a\sin\theta} r\left(\frac{a^2-r^2}{a}\right) dr\, d\theta$ $=\int_0^{\pi/2}\left[\int_0^{a\sin\theta}\left(ar-\frac{1}{a}r^3\right) dr\right] d\theta$ $=\int_0^{\pi/2}\left[\frac{a}{2}r^2-\frac{1}{4a}r^4\right]_0^{a\sin\theta} d\theta$ $=\int_0^{\pi/2}\left(\frac{a^3}{2}\sin^2\theta-\frac{a^3}{4}\sin^4\theta\right) d\theta$ $=\frac{a^3}{2}\cdot\frac{1}{2}\cdot\frac{\pi}{2}-\frac{a^3}{4}\cdot\frac{3\cdot1}{4\cdot2}\cdot\frac{\pi}{2}$ $=\frac{\pi a^3}{8}-\frac{3\pi a^3}{64}=\frac{5\pi a^3}{64}$. **(Ans.)**						
26	Unit 5	Find the volume of the tetrahedron bounded by the coordinate planes and the planes $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$	12	4	Mediu m	Difficult	CO-3	Numerical

Solution. The required volume OABC lies between the plane $z = c\left(1 - \frac{x}{a} - \frac{y}{b}\right)$ and is bounded on the sides by the planes $x = 0$, $y = 0$ and $z = 0$.

Limits of z : $z = 0$ and $z = c\left(1 - \frac{x}{a} - \frac{y}{b}\right)$

Limits of y : $y = 0$ and $y = b\left(1 - \frac{x}{a}\right)$

Limits of x : $x = 0$ and $x = a$



Volume $= \int\int\int dx\,dy\,dz$

$$= \int_0^a dx \int_0^{b\left(1-\frac{x}{a}\right)} dy \int_0^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)} dz$$

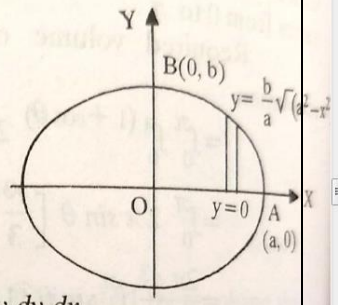
$$= \int_0^a dx \int_0^{b\left(1-\frac{x}{a}\right)} dy \,[z]_0^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)}$$

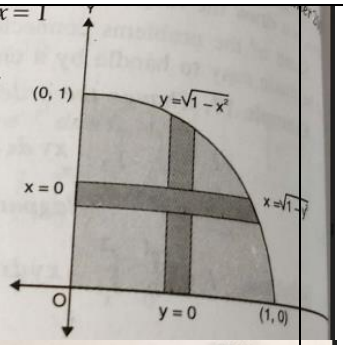
$$= \int_0^a dx \int_0^{b\left(1-\frac{x}{a}\right)} c\left(1 - \frac{x}{a} - \frac{y}{b}\right) dy$$

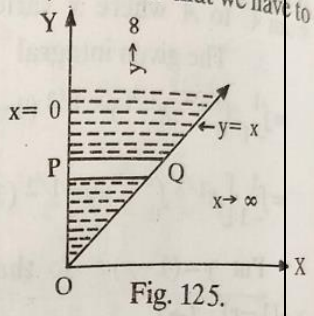
$$= c\int_0^a dx \left[\left(1-\frac{x}{a}\right)y - \frac{y^2}{2b}\right]_0^{b\left(1-\frac{x}{a}\right)}$$

$$= c\int_0^a \left[b\left(1-\frac{x}{a}\right)^2 - \frac{b}{2}\left(1-\frac{x}{a}\right)^2\right] dx$$

		$= \frac{bc}{2}\int_0^a \left(1-\frac{x}{a}\right)^2 dx$ $= \frac{bc}{2}\left[\frac{\left(1-\frac{x}{a}\right)^3}{-\frac{3}{a}}\right]_0^a$ $= -\frac{abc}{6}(0-1) = \frac{1}{6}abc.$						
27	Unit 5	Find the mass of the ellipse plate $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, if the density at any point (x,y)on it is μxy	15	4	High	Very difficult	CO-3	Numerical

		Solution. Let the required mass be M which is four times the mass in the first quadrant. For the region OAB, x varies from 0 to a and y varies from 0 to $\frac{b}{a}\sqrt{(a^2-x^2)}$.  Fig. 153. $\therefore \quad M=4\int_0^a \int_0^{(b/a)\sqrt{(a^2-x^2)}} \mu\, xy\, dy\, dx$ $=4\int_0^a \mu x \left[\frac{y^2}{2}\right]_0^{(b/a)\sqrt{(a^2-x^2)}} dx$ $=2\mu\int_0^a x\cdot\frac{b^2}{a^2}(a^2-x^2)\,dx$ $=2\mu\cdot\frac{b^2}{a^2}\int_0^a (a^2x-x^3)\,dx$ $=2\mu\cdot\frac{b^2}{a^2}\left[a^2\cdot\frac{x^2}{2}-\frac{x^4}{4}\right]_0^a$ $=2\mu\cdot\frac{b^2}{a^2}\times\frac{a^4}{4}=\frac{\mu a^2b^2}{2}$. (Ans.)						
28	Unit 5	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2\,dydx$		4	High	Very difficult	CO-3	Numerical

		Solution. The region of integration is from $x = 0$ to $x = 1$ and $y = 0$ to $y = \sqrt{1-x^2}$. Now change the order of integration with respect to x from $x = 0$ to $x = \sqrt{1-y^2}$ and then $y = 0$ to $y = 1$. $\therefore \int_0^1 \int_0^{\sqrt{1-x^2}} y^2\,dy\,dx$ $= \int_0^1 \int_0^{\sqrt{1-y^2}} y^2\,dx\,dy = \int_0^1 y^2 [x]_0^{\sqrt{1-y^2}}\,dy$  137 $= \int_0^1 y^2\sqrt{1-y^2}\,dy$ [Put $y = \sin\theta$ so that $dy = \cos\theta\,d\theta$; when $y = 0$, then $\theta = 0$ and $y = 1$, then $\theta = \frac{\pi}{2}$] $= \int_0^{\frac{\pi}{2}} \sin^2\theta\cos\theta\cos\theta\,d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{2\sin\theta\cos\theta}{2}\right)^2 d\theta$ $= \frac{1}{4}\int_0^{\frac{\pi}{2}} \sin^2 2\theta\,d\theta = \frac{1}{8}\int_0^{\frac{\pi}{2}} (1-\cos 4\theta)\,d\theta$ $= \frac{1}{8}\left(\theta - \frac{\sin 4\theta}{4}\right)_0^{\frac{\pi}{2}}$ $= \frac{1}{8}\left\{\left(\frac{\pi}{2}-0\right) - \frac{1}{4}(\sin 2\pi - \sin 0)\right\} = \frac{\pi}{16}$. Ans						
29	Unit 5	Evaluate: $\int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}dydx$ by changing into polar coordinates		4	High	Very difficult	CO-3	Numerical

		Solution. From the limits of integration, it is clear that we have to integrate first w.r.t. y which varies from $y=x$ to $y=\infty$ and then w.r.t. x which varies from $x=0$ to $x=\infty$. The region of integration is shown shaded in figure. To change the order of integration we means to integrate first w.r.t. x and then w.r.t. y.  Fig. 125. To cover the region of integration, we draw a strip PQ parallel to X-axis where x varies from $x=0$ to $x=y$. Then this strip slides from O to infinite space where y varies from $y=0$ to $y=\infty$. $\therefore \int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\,dy\,dx = \int_0^\infty \int_0^y \frac{e^{-y}}{y}\,dy\,dx$ $= \int_0^\infty \left[\frac{e^{-y}}{y}\int_0^y dx\right] dy$ $= \int_0^\infty \frac{e^{-y}}{y}\left[x\right]_0^y dy$ $= \int_0^\infty \frac{e^{-y}}{y}\cdot y\,dy$ $= \int_0^\infty e^{-y}\,dx$ $= \left[\frac{e^{-y}}{-1}\right]_0^\infty = [0+1] \quad \because \lim_{y\to\infty} e^{-y}=0$ $=1.$ (Ans.)						
30	Unit 5	Evaluate: $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\,dx\,dy\,dz$	15	4	High	Very difficult	CO-3	Numerical

(S. 2008)

Solution. We have to integrate first w.r.t. z, then w.r.t. y and finally w.r.t. x.

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dx\, dy\, dz$$

$$= \int_0^1 \left\{ \int_0^{\sqrt{1-x^2}} xy \left[\int_0^{\sqrt{1-x^2-y^2}} z\, dz \right] dy \right\} dx$$

$$= \int_0^1 \left\{ \int_0^{\sqrt{1-x^2}} xy \left[\frac{z^2}{2} \right]_0^{\sqrt{1-x^2-y^2}} dy \right\} dx$$

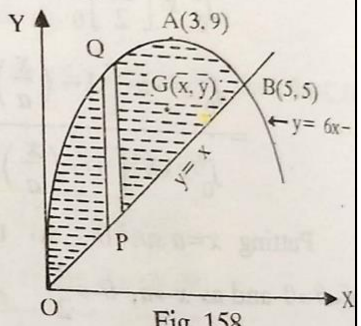
$$= \frac{1}{2} \int_0^1 \left\{ \int_0^{\sqrt{1-x^2}} xy \left(1-x^2-y^2\right) dy \right\} dx$$

$$= \frac{1}{2} \int_0^1 x \left[\left(1-x^2\right) \frac{y^2}{2} - \frac{y^4}{4} \right]_0^{\sqrt{1-x^2}} dx,$$

by parts

$$= \frac{1}{2} \int_0^1 x \left[\frac{\left(1-x^2\right)^2}{2} - \frac{\left(1-x^2\right)^2}{4} \right] dx$$

$$= \frac{1}{2} \int_0^1 x \cdot \frac{\left(1-x^2\right)^2}{4} dx$$

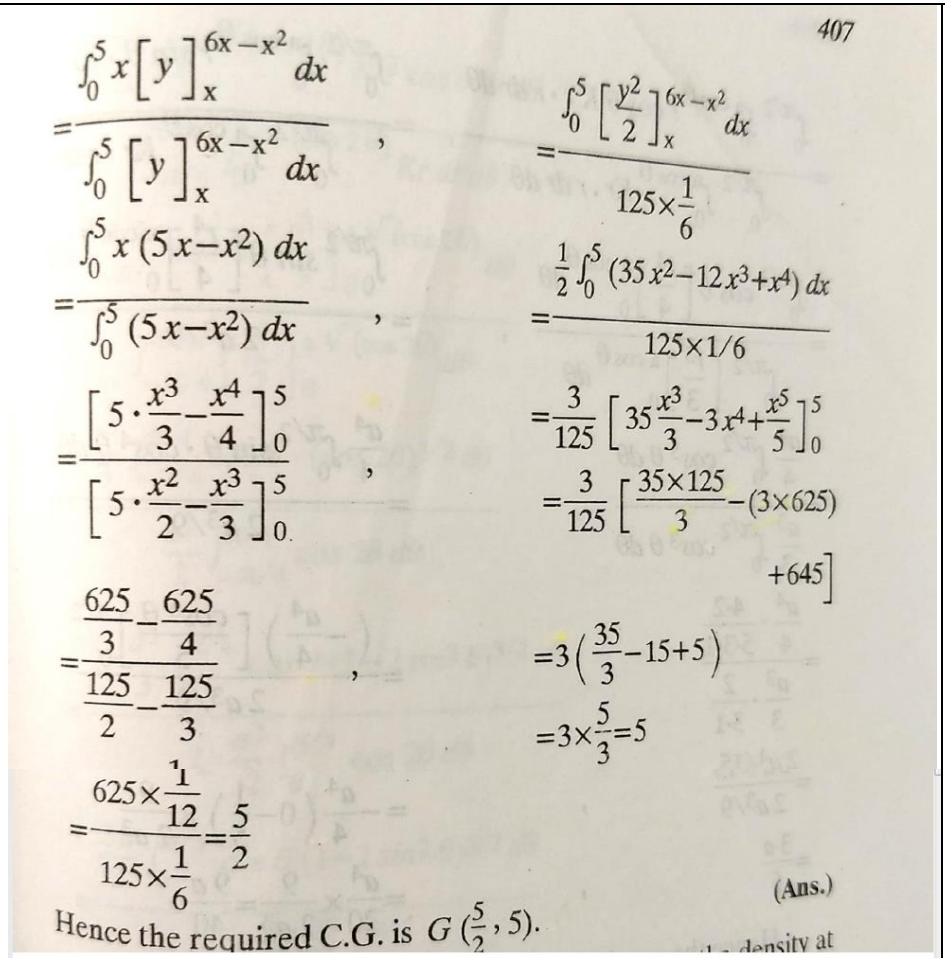
31	Unit 5	Find the center of gravity of the area between $y = 6x - x^2$ and $y = x$ **Solution.** The equation of the curve can be written as $x^2 - 6x = -y$ $\Rightarrow x^2 - 6x + 9 = 9 - y$ $\Rightarrow (x-3)^2 = -(y-9)$ $\therefore$ the curve is a parabola having its vertex at $A(3, 9)$ and it i opening downwards. To find the points of intersection of the parabola and the line, w equate the values of y from their equations. $x = 6x - x^2$ $\Rightarrow x^2 - 5x = 0$ $\Rightarrow x(x-5) = 0$ $\Rightarrow x = 0$ or $x = 5$ when $x - 0$, $y = 0$ and when $x = 5$, $y = 5$. $\therefore$ The points of intersection are $O(0, 0)$ and $B(5, 5)$ The area A is shown shaded.  Fig. 158. For this area y varies from x to $6x - x^2$ and x varies from 0 to Since the density is not given but it is [redacted] onstant	15	4	High	Very difficult	CO-3	Numerical

For this area y varies from x to $6x-x^2$ and x varies from 0 to 5.

Since the density is not given, let it be K, a constant.

If $G(\bar{x}, \bar{y})$ is the C.G. of the area, then

$$\bar{x} = \frac{\int_0^5 \int_x^{6x-x^2} x\,K\,dy\,dx}{\int_0^5 \int_x^{6x-x^2} K\,dy\,dx}, \qquad \bar{y} = \frac{\int_0^5 \int_x^{6x-x^2} y\,K\,dy\,dx}{\int_0^5 \int_x^{6x-x^2} K\,dy\,dx}$$

								
32	Unit 5	Evaluate $\int_0^{\log 2}\int_0^{x}\int_0^{x+\log y} e^{x+y+z}\,dzdydx$	15	4	High	Very difficult	CO-3	Numerical

		Solution. First we are to integrate w.r.t. z, then w.r.t. y and finally w.r.t. x. $I=\int_0^{\log 2}\int_0^{x}\left[\int_0^{x+\log y} e^{x+y+z}\,dz\right]dy\,dx$ $=\int_0^{\log 2}\int_0^{x}\left[e^{x+y+z}\right]_0^{x+\log y}dy\,dx$ $=\int_0^{\log 2}\int_0^{x}\left[e^{x+y+x+\log y}-e^{x+y}\right]dy\,dx$ $=\int_0^{\log 2}\left[\int_0^{x}\left(e^{2x+y}.y-e^{x+y}\right)dy\right]dx$ $=\int_0^{\log 2}\left[y\,e^{2x+y}-e^{2x+y}-e^{x+y}\right]_0^{x}dx$ $=\int_0^{\log 2}[x\,e^{3x}.-e^{3x}-e^{2x}+e^{2x}+e^{x}]\,dx$ $=\int_0^{\log 2}[(x-1)\,e^{3x}+e^{x}]\,dx$ $=\left[(x-1)\frac{e^{3x}}{3}-\frac{e^{3x}}{9}+e^{x}\right]_0^{\log 2}$ $=(\log 2-1)\cdot\frac{8}{3}-\frac{8}{9}+2+\frac{1}{3}+\frac{1}{9}-1$ $=\frac{8}{3}\log 2-\frac{8}{3}-\frac{8}{9}+1+\frac{1}{3}+\frac{1}{9}$ $=\frac{8}{3}\log 2+\frac{-24-8+9+3+1}{9}$ (Ans.) $=\frac{8}{3}\log 2-\frac{19}{9}$.						
33	Unit 5	Find the center of gravity of the area bounded by parabola $y^2=x$, and the line X+Y= 2	15	4	High	Very difficult	CO-3	Numerical

Points of intersection of (1) and (2)

From (2), $x=2-y$. Putting in (1), we get

$$y^2=2-y \Rightarrow y^2+y-2=0$$

$$\Rightarrow (y+2)(y-1)=0 \Rightarrow y=-2, 1.$$

When $y=-2$, (2) gives $x=4$; and when $y=1$, (2) gives $x=1$,

$\therefore$ points of intersection of (1) and (2) are $(4,-2)$ and $(1,1)$.

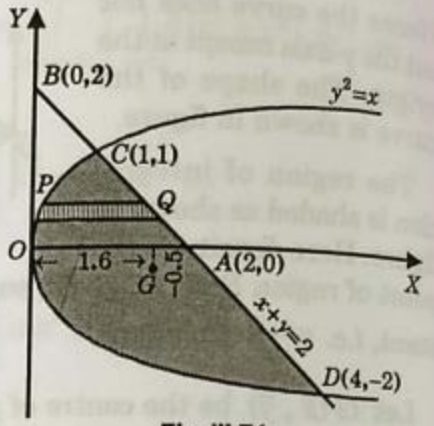


Fig. III.74

The region of integration is shaded as shown in figure.

Here density at the point of a region is not given, so consider it as constant, *i.e.* $\rho=k$ (constant)

Let $G(\bar{x}, \bar{y})$ be the C. G. of a region. Then

$$\bar{x}=\frac{\iint_R \rho\, x\, dx\, dy}{\iint_R \rho\, dx\, dy}, \quad \bar{y}=\frac{\iint_R \rho\, y\, dx\, dy}{\iint_R \rho\, dx\, dy}$$

Suppose we have to integrate first w.r.t. x and then w.r.t. y. Draw a strip PQ parallel to x-axis where x varies from $x=y^2$ to $x=2-y$ and strip PQ is then slides from a lower extreme D to an upper extreme C where y varies from $y=-2$ to $y=1$.

34	Unit 5	Find by double integration the area outside the circle r =a$\cos\theta$ and inside the circle r = 2a $\cos\theta$	15	4	High	Very difficult	CO-3	Numerical

Solution. Given curves are

ellipse : $4x^2+9y^2=36 \Rightarrow \frac{x^2}{9}+\frac{y^2}{4}=1$ ...(1)

line : $2x+3y=6$ ...(2)

Point of intersection of (1) and (2).

From (2) $3y=6-2x$, using in (1), we get

$$4x^2+(6-2x)^2=36$$

$$\Rightarrow 4x^2+36-24x+4x^2=36$$

$$\Rightarrow 8x^2-24x=0$$

$$\Rightarrow x(8x-24)=0$$

$$\Rightarrow x=0,3$$

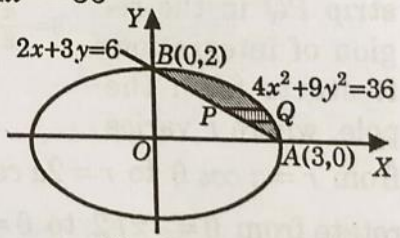


Fig. III.51

when $x=0$, (2) gives $y=2$;

and when $x=3$, (2) gives $y=0$

$\therefore$ points of intersection of (1) and (2) are $A(3, 0)$ and $B(0, 2)$. The region of integration is shaded as shown in figure.

Suppose we have to integrate first w.r.t. x and then

w.r.t. y. Draw a strip PQ parallel to x-axis in the region of integration where x varies from $x=\frac{6-3y}{2}$

$=\frac{3}{2}(2-y)$ to $x=\sqrt{\frac{36-9y^2}{4}}=\frac{3}{2}\sqrt{4-y^2}$ and strip PQ is then slides from A to B where y varies from $y=0$ to $y=2$,

$$\therefore \text{ required area } = \int_{y=0}^{2} \int_{x=3/2\,(2-y)}^{3/2\,\sqrt{4-y^2}} dx\,dy$$

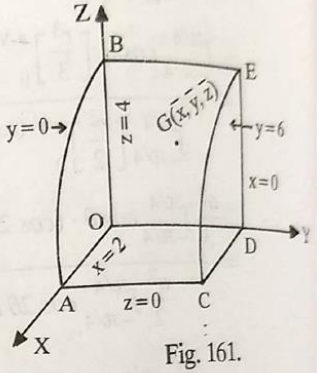
$$= \int_0^2 \left[x\right]_{3/2\,(2-y)}^{3/2\,\sqrt{4-y^2}} dy$$

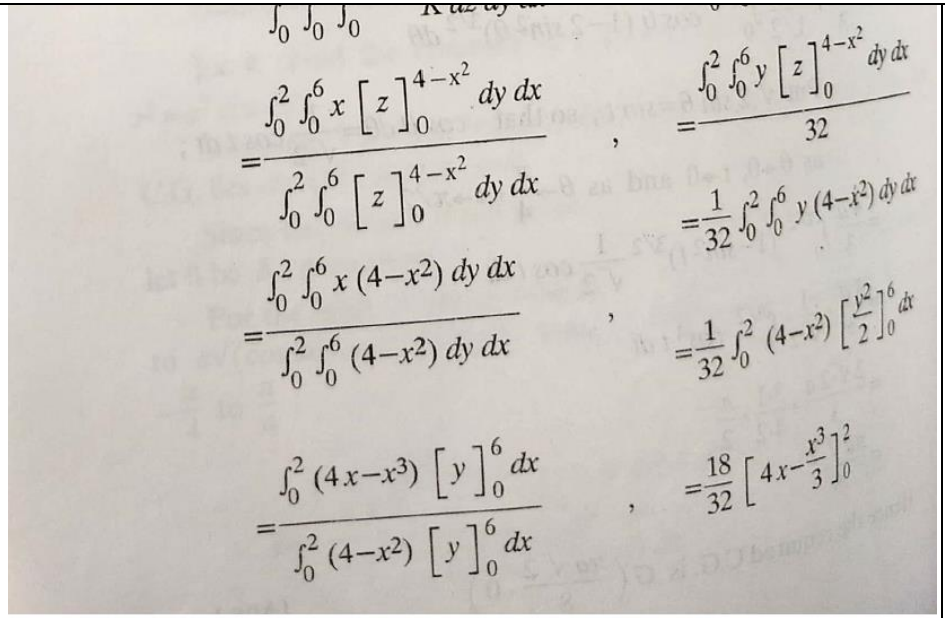
$$= \int_0^2 \left[\frac{3}{2}\sqrt{4-y^2} - \frac{3}{2}(2-y)\right] dy$$

$$= \frac{3}{2}\left[\frac{y\sqrt{4-y^2}}{2} + \frac{4}{2}\sin^{-1}\frac{y}{2} - 2y + \frac{y^2}{2}\right]_0^2$$

$$= \frac{3}{2}\left[2\,\frac{\pi}{2} - 4 + 2\right] = \frac{3}{2}(\pi-2) \text{ sq. units.}$$

(Ans.)

35	Unit 5	Find the centroid of the region R bounded by the parabolic cylinder $z = 4 - x^2$, and the planes $x = 0, y = 0, y = 6, z = 0$. **Solution.** The given solid is parabolic cylinder whose axis is parallel to Y-axis. Since the density is not given, let it be K, a constant. We have to consider the portion OAB, ECD of the given solid as shown in the adjoining figure. To cover the whole volume x varies from 0 to 2 y varies from 0 to 6 and z varies from 0 to $4-x^2$.  Fig. 161. If $G(\bar{x}, \bar{y}, \bar{z})$ is the C.G. of the solid, then $\bar{x} = \dfrac{\int_0^2 \int_0^6 \int_0^{4-x^2} x \cdot \rho \, dz \, dy \, dx}{\int_0^2 \int_0^6 \int_0^{4-x^2} \rho \, dz \, dy \, dx}$, $\bar{y} = \dfrac{\int_0^2 \int_0^6 \int_0^{4-x^2} y \cdot \rho \, dz \, dy \, dx}{\int_0^2 \int_0^6 \int_0^{4-x^2} \rho \, dz \, dy \, dx}$ $= \dfrac{\int_0^2 \int_0^6 \int_0^{4-x^2} K x \, dz \, dy \, dx}{\int_0^2 \int_0^6 \int_0^{4-x^2} K \, dz \, dy \, dx}$, $= \dfrac{\int_0^2 \int_0^6 \int_0^{4-x^2} K y \, dz \, dy \, dx}{\int_0^2 \int_0^6 \int_0^{4-x^2} K \, dz \, dy \, dx}$	15	4	High	Very difficult	CO-3	Numerical



$$\int_0^2 \int_0^6 \int_0$$

$$= \frac{\int_0^2 \int_0^6 x \left[z \right]_0^{4-x^2} dy\, dx}{\int_0^2 \int_0^6 \left[z \right]_0^{4-x^2} dy\, dx}$$

$$= \frac{\int_0^2 \int_0^6 x\,(4-x^2)\, dy\, dx}{\int_0^2 \int_0^6 (4-x^2)\, dy\, dx}$$

$$= \frac{\int_0^2 (4x-x^3) \left[y \right]_0^6 dx}{\int_0^2 (4-x^2) \left[y \right]_0^6 dx}$$

$$= \frac{\int_0^2 \int_0^6 y \left[z \right]_0^{4-x^2} dy\, dx}{32}$$

$$= \frac{1}{32} \int_0^2 \int_0^6 y\,(4-x^2)\, dy\, dx$$

$$= \frac{1}{32} \int_0^2 (4-x^2) \left[\frac{y^2}{2} \right]_0^6 dx$$

$$= \frac{18}{32} \left[4x - \frac{x^3}{3} \right]_0^2$$

$$=\frac{6\left[2x^2-\frac{x^4}{4}\right]_0^2}{6\left[4x-\frac{x^3}{3}\right]_0^2}, \quad =\frac{9}{16}\times\frac{16}{3}=3$$

$$=\frac{6\times4}{6\times\frac{16}{3}}=\frac{3}{4}$$

$$\bar{z}=\frac{\int_0^2\int_0^6\int_0^{4-x^2} Kz\,dz\,dy\,dx}{\int_0^2\int_0^6\int_0^{4-x^2} K\,dz\,dy\,dx}$$

$$=\frac{\int_0^2\int_0^6\left[\frac{z^2}{2}\right]_0^{4-x^2} dy\,dx}{32}$$

$$=\frac{1}{32}\int_0^2\int_0^6 \frac{1}{2}(16-8x^2+x^4)\,dy\,dx$$

$$= \frac{1}{32} \int_0^2 \int_0^6 \frac{1}{2} (16 - 8x^2 + x^4)\, dy\, dx$$

$$= \frac{1}{32} \int_0^2 \frac{1}{2} (16 - 8x^2 + x^4) \left[y \right]_0^6 dx$$

$$= \frac{1}{32} \int_0^2 3 (16 - 8x^2 + x^4)\, dx$$

$$= \frac{3}{32} \left[16x - \frac{8}{3}x^3 + \frac{x^5}{5} \right]_0^2$$

$$= \frac{3}{32} \left[32 - \frac{64}{3} + \frac{32}{5} \right]$$

$$= 3\left[1 - \frac{2}{3} + \frac{1}{5} \right] = 3 \times \frac{8}{15} = \frac{8}{5}$$

(An

Hence the required C.G. is $G\left(\frac{3}{4}, 3, \frac{8}{5}\right)$.

Practice

the area